The Saturday Mornings.tv

TODAY'S FORECAST
LIGHT DRIZZLE OF COMMANDS,
SUNNY BY TEATIME

THE FREE EDITION
STATION MANUAL
BACK TO THE FRONT PAGE

DO-IT-YOURSELF SUPPLEMENT · PART ONE SATURDAYMORNINGS.TV

"Some assembly required"

THE FREE EDITION · STEP BY STEP

BUILD YOUR OWN STATION

The parent app and the server that runs it, from a bare computer to a working station on your own free Cloudflare account. It covers every account, key and command. Put the kettle on.

27 STEPS | 4 ACCOUNTS | 1 AFTERNOON | $0 ON FREE TIERS

Difficulty, in bowls of cereal: 🥣 easy · 🥣🥣 fiddly · 🥣🥣🥣 pour yourself another bowl

NEXT: SAMSUNG TV GUIDE ANDROID TV GUIDE THE SHORT VERSION FRONT PAGE

WHAT YOU'LL NEED

- [ ] **A computer** (Mac, Windows or Linux) where you can install things and type commands into a terminal (the black window with the blinking cursor).
- [ ] **A Cloudflare account**, free plan. Your station's server and database live here.
- [ ] **A Google account** for a Google Cloud project, which hands out the YouTube key.
- [ ] **A Resend account**, free. It emails the 6-digit sign-in codes.
- [ ] **A phone** for the parent app (iPhone or Android).

- [ ] **Optional: a domain** you control, like `your-domain.com`. Passkeys and email work much better with one.
- [ ] **A TV** for later. The Samsung and Android TV steps have their own guides.
- [ ] **The code**, free on GitHub at github.com/RonzGore/saturday-mornings.

TIME, FIRST GO	WAITING FOR DNS	COST	DOMAIN (OPTIONAL)
3 to 4 hours	**minutes, or a day**	**$0 on free tiers**	**about $10 a year**

PART 1 · THE TOOLS *Set up your computer*

1

Open a terminal

Everything below that sits in a dark box is typed into a terminal. On a Mac, open **Terminal** (press Cmd + Space and type "Terminal"). On Windows, open **PowerShell**. On Linux, you already know.

Tip: each dark box has a **Copy** button. Paste, press Return, read what it says.

2 MIN

2

Install Node.js (with npm)

Node.js runs the build tools; npm, which comes with it, fetches them. Download the **LTS** version from `nodejs.org` and install it like any other app. You need version 20 or newer.

TYPE THIS

```
node -v
npm -v
```

YOU SHOULD SEE: a version like `v22.11.0` (anything from v20 up) and a second number for npm.

10 MIN

3

Get the code

Copy the code to your computer with `git`. If you'd rather not use git, open github.com/RonzGore/saturday-mornings, click the green **Code** button, choose **Download ZIP** and unzip it.

TYPE THIS

```
git clone https://github.com/RonzGore/saturday-mornings.git
cd saturday-mornings
```

YOU SHOULD SEE: a folder called `saturday-mornings` with `apps`, `cloud`, `shared` and `home-server` inside it. The phone app is in `apps/parent`, the TV screens in `apps/tv`, the Samsung app in `apps/tv-samsung`.

5 MIN

4

Install the station's parts

The server lives in the `cloud` folder. This downloads what it needs, including **Wrangler**, Cloudflare's command-line tool. You'll run it as `npx wrangler`.

TYPE THIS

```
cd cloud
npm install
```

YOU SHOULD SEE: `added … packages` and no red `ERR!` lines. Warnings in yellow are fine.

From here on, every command runs inside `saturday-mornings/cloud` unless a step says otherwise.

5 MIN

PART 2 · THE ACCOUNTS *Four sign-ups and two keys*

5

Open a Cloudflare account

Sign up at `dash.cloudflare.com/sign-up` and verify your email. The **Free** plan is plenty for a household. No card needed.

YOU SHOULD SEE: the Cloudflare dashboard, with “Workers & Pages” in the left-hand menu.

10 MIN

6

Log Wrangler in to Cloudflare

This opens your browser and asks Cloudflare to trust your computer. Click **Allow**, then come back to the terminal.

TYPE THIS

```
npx wrangler login
npx wrangler whoami
```

YOU SHOULD SEE: "Successfully logged in", then your account email and account name in a little table.

3 MIN

7

Make a Google Cloud project

YouTube lookups need a key from Google. Go to `console.cloud.google.com`, sign in, and use the project picker at the top to make a **New project** (call it "saturday-mornings").

Then open **APIs & Services › Library**, search for **YouTube Data API v3** (not "YouTube Analytics", not "YouTube Reporting") and press **Enable**.

YOU SHOULD SEE: the YouTube Data API v3 page with a "Manage" button and the word "Enabled".

10 MIN

8

TRICKY!

Create the YouTube key, and fence it in

In **APIs & Services › Credentials**, choose **Create credentials › API key**. Copy the key somewhere safe for step 16.

Then click the key to edit it. Under **API restrictions** choose **Restrict key** and tick only **YouTube Data API v3**. Leave **Application restrictions** at None, because your server calls YouTube rather than a website, and a website restriction would block it. Save.

YOU SHOULD SEE: your key listed with a green tick and "1 API" under restrictions.

A restricted key is a key that is only good for one door. If it ever leaks, nobody can use it to run up anything else on your Google account.

10 MIN

9

Learn the YouTube quota

Google gives each project **10,000 units a day**, free. A search costs 100 units; looking up a channel, playlist or video costs 1.

Saturday Mornings is careful with them. It caps search at 30 a day per parent and 80 a day in total, and remembers results for 6 hours. If you do run out, the app says “Search is resting until tomorrow”, and pasting links still works.

> **YOU SHOULD SEE:** nothing to do. Just don't let a four-year-old near the search box with a stopwatch.

2 MIN

10

Open a Resend account and make a key

Resend sends the sign-in emails. Sign up free at `resend.com`, then go to **API Keys › Create API key** with **Sending access**. Copy it (it starts with `re_`) for step 16.

Until you verify a domain (step 23), Resend sends from its test address, `onboarding@resend.dev`*, and only to* **the email you signed up to Resend with**. *Fine for a station of one; the other parent needs step 23.*

> **YOU SHOULD SEE:** your new key in the API Keys list (Resend only shows the full key once, so copy it now).

10 MIN

PART 3 · THE SERVER *Build it, fill it, switch it on*

11

Create the database

Your family's kids, channels, shows and times live in a **D1** database (Cloudflare's small, free database). Keep the name exactly `saturday-morning`, because the deploy script looks for it by that name.

TYPE THIS

```
npx wrangler d1 create saturday-morning
```

> **YOU SHOULD SEE:** “Successfully created DB” and a `database_id`, a long string like `1a2b3c4d-…`. Copy it. (If Wrangler asks whether to add it to your config for you, say no, because you'll do it by hand in the next step.)

3 MIN

12

Paste your database id

Open `cloud/wrangler.jsonc` in any text editor (TextEdit in plain-text mode, Notepad, or VS Code). It's the station's settings file. Find `d1_databases` and replace the id that's there with yours.

IN CLOUD/WRANGLER.JSONC

```
"d1_databases": [
  {
    "binding": "DB",
    "database_name": "saturday-morning",
    "database_id": "PASTE-YOUR-DATABASE-ID-HERE",
    "migrations_dir": "migrations"
  }
],
```

YOU SHOULD SEE: your own id between the quotes, and every quote and comma still in place.

3 MIN

13

TRICKY!

Replace our settings with yours

The same file is full of *our* station's addresses. Change these, and keep the quotes. Lines starting with `//` are notes for humans; the server ignores them.

- `routes`: delete this whole line for now. It points at our address; you'll add your own in step 21.
- `workers_dev`: leave it `true`. That gives you a free `workers.dev` address.
- `APP_NAME`: the name in emails and on screen. “Saturday Mornings” is fine, or name it after your house.
- `RP_ID`: the passkey domain. Delete this line for now; you'll set it to your own domain in step 22. (Without it, passkeys use whatever address the station is on.)
- `EMAIL_FROM`: who sign-in emails are from. Until step 23, use Resend's test sender as below.
- `REPLY_TO`: where replies to waiting-list emails go. Your own email.
- `EARLY_ACCESS`: `"off"` for a family station (see the box below), or delete the line.
- `ADMIN_EMAILS`: your email, and the other parent's if you like, separated by commas. Admins see **Account › Early access** and can always sign in.

IN CLOUD/WRANGLER.JSONC: YOUR VARS SHOULD LOOK LIKE THIS

```
"workers_dev": true,
"vars": {
  "APP_NAME": "Saturday Mornings",
  "EMAIL_FROM": "Saturday Mornings <onboarding@resend.dev>",
```

```
  "REPLY_TO": "you@example.com",
  "EARLY_ACCESS": "off",
  "ADMIN_EMAILS": "you@example.com,partner@example.com"
},
```

YOU SHOULD SEE: no `saturdaymornings.tv` addresses and no email of ours left anywhere in the file. Search it for "saturdaymornings" to be sure.

Leave `triggers` alone. The daily one (`17 9 * * *`, 09:17 UTC) keeps YouTube titles fresh; the every-minute one thanks people who join the waiting list and does nothing on a family station.

15 MIN

Who can sign in to your station?

This is the `EARLY_ACCESS` switch. Pick one:

"OFF" (OR LEFT OUT)
Anyone who can reach your station's address can create an account. Simple, and fine as long as you keep the address to yourselves, like a house key.

"ON"
Only approved emails can sign in; everyone else joins a waiting list. Admins (`ADMIN_EMAILS`) always get in. Let family in from **Account › Early access › Let someone in by email**. (Untick "Send them the You're in email": that email is written for our hosted service.)

Either way, the other parent can always join through a family invite (step 26).

14

Put your contact email on the legal pages

Your station has Privacy and Terms pages, and a help link in the app. In `cloud/wrangler.jsonc`, set `CONTACT_EMAIL` inside `"vars"` to yours.

IN CLOUD/WRANGLER.JSONC, INSIDE "VARS"

```
"CONTACT_EMAIL": "you@example.com",
```

YOU SHOULD SEE: your email between the quotes. You'll see it on `/privacy` after deploying.

2 MIN

15

Invent a secret “pepper”

A long random string the server mixes into sign-in codes so they can't be guessed. Nobody ever types it. Let the computer make one (on Mac or Linux):

TYPE THIS

```
openssl rand -hex 32
```

YOU SHOULD SEE: 64 random letters and numbers. Copy them for the next step. (On Windows, mash the keyboard for 40 characters. Not `password123`. Not the dog's name.)

2 MIN

16

Hand the station its secrets

Secrets are settings too private for the settings file. Cloudflare stores them, locked. Run these one at a time; each one asks you to paste a value and press Return. (The first time, Wrangler may offer to create the Worker for you; say yes.)

TYPE THIS, ONE LINE AT A TIME

```
npx wrangler secret put CODE_PEPPER      # paste the pepper from step 15
npx wrangler secret put RESEND_API_KEY   # paste the re_ key from step 10
npx wrangler secret put YOUTUBE_API_KEY  # paste the Google key from step 8
```

YOU SHOULD SEE: “Success! Uploaded secret CODE_PEPPER”, and the same for the other two.

Skip one and things go quiet. Without `RESEND_API_KEY` *nobody can sign in, and without* `YOUTUBE_API_KEY` *channel links and search fail (single videos and playlists still work).*

5 MIN

17

WOO!

Deploy!

One command builds the parent app, the TV app and the sign-in pages, sets up the database tables and publishes your station.

TYPE THIS

```
npm run deploy
```

YOU SHOULD SEE: migrations listed with ✅, then “Deployed saturday-morning” and your address:

`https://saturday-morning.YOUR-SUBDOMAIN.workers.dev`. Write it down.

5 MIN

18

Is anybody home?

The health check is a tiny page that answers if the station is up. Open it in a browser, or ask from the terminal:

TYPE THIS (YOUR ADDRESS)

```
curl https://saturday-morning.YOUR-SUBDOMAIN.workers.dev/api/health
```

YOU SHOULD SEE: `{"ok":true}`. If the address isn't found yet, give it a minute, because brand new `workers.dev` addresses can take a moment to appear.

2 MIN

19

OPTIONAL

Want to try it on your computer first?

You can run the whole station on your own computer before (or instead of) deploying. Make a file called `cloud/.dev.vars` with the line below, and sign-in codes appear on screen instead of being emailed. You can add a line `YOUTUBE_API_KEY=YOUR_YOUTUBE_KEY` too. This file stays on your computer, so never share it.

IN CLOUD/.DEV.VARS

```
DEV_ECHO_CODES=1
```

TYPE THIS

```
npm run dev
```

YOU SHOULD SEE: "Ready on http://localhost:8787". Open it in your browser and sign in with any email; the code shows on the page.

10 MIN

PART 4 · A PROPER ADDRESS *Optional, in the way seatbelts are optional*

20

RECOMMENDED

Put a domain on Cloudflare

Your own address, like `tv.your-domain.com`, makes passkeys stick, lets email reach the whole family, and lets thumbnails load faster (picture caching only works on a custom domain). Buy one from Cloudflare (**Domain Registration**), or bring one you own: **Add a domain** in the dashboard, then change your registrar's nameservers to the two Cloudflare gives you.

> **YOU SHOULD SEE:** your domain marked **ACTIVE** in the Cloudflare dashboard. That can take minutes, or a day. Nobody knows which.

15 MIN + WAITING

21

Move the station to your address

Add a `routes` line back to `cloud/wrangler.jsonc` with your address, then deploy again. Cloudflare makes the certificate (the padlock) for you.

IN CLOUD/WRANGLER.JSONC

```
"routes": [{ "pattern": "tv.your-domain.com", "custom_domain": true }],
"workers_dev": true,
```

TYPE THIS

```
npm run deploy
```

> **YOU SHOULD SEE:** `{"ok":true}` at `https://tv.your-domain.com/api/health`. The padlock can take a few minutes, occasionally longer.

10 MIN

22

Tie passkeys to your domain

A passkey lets you sign in with your face or fingerprint instead of an email code. Passkeys only work on the address they were made on, so set `RP_ID` now, before anyone makes one. Use your main domain, so passkeys keep working if the station moves to another `something.your-domain.com`.

IN CLOUD/WRANGLER.JSONC, INSIDE "VARS"

```
"RP_ID": "your-domain.com",
```

> **YOU SHOULD SEE:** after `npm run deploy`, nothing different yet. You'll make your first passkey in step 24. Made one on `workers.dev` already? Make a fresh one on the new address. Email codes always work.

3 MIN

23

TRICKY!

Teach email to be trusted

So codes reach the other parent too. In Resend, **Domains** › **Add domain**: use a sending subdomain like `send.your-domain.com`. Resend shows a few DNS records (lines of settings that tell the world your domain allows it). Add each one in Cloudflare under your domain's **DNS** › **Records**:

- **MX** on `send` (where bounced mail goes)
- **TXT** on `send`, the SPF record (who may send)
- **TXT** on `resend._domainkey.send`, the DKIM key (a signature)
- Optional: **TXT** on `_dmarc` (what to do with fakes)

Copy the values exactly from Resend. When it says **Verified**, change `EMAIL_FROM` and deploy:

IN CLOUD/WRANGLER.JSONC, INSIDE "VARS"

```
"EMAIL_FROM": "Saturday Mornings <hello@send.your-domain.com>",
```

TYPE THIS

```
npm run deploy
```

YOU SHOULD SEE: green “Verified” in Resend, and sign-in codes arriving for any email. If they don't, run `npx wrangler tail` and sign in again; problems show up as `resend failed` and a number.

30 MIN + WAITING

PART 5 · ON THE AIR *Sign in, set up, switch on the telly*

24

Sign in for the first time

On your phone, open your station's address (`https://tv.your-domain.com`, or the `workers.dev` one). Type your email, then the 6-digit code from your inbox. Then, if you like, choose **Add a passkey on this device** so next time is one tap.

YOU SHOULD SEE: the parent app, empty and waiting for its first kid. No code arriving? Check spam, then step 10's catch, then the Stuck box below.

3 MIN

25

Put the parent app on your home screen

It's a web app, so there's no app store. It installs straight from the browser and opens full screen like any other app.

- **iPhone:** in **Safari**, tap **Share** (the square with the arrow), then **Add to Home Screen**, then **Add**.
- **Android:** in **Chrome**, tap ⋮ (top right), then **Install app** (on some phones, **Add to Home screen**).

> **YOU SHOULD SEE:** the Saturday Mornings icon on your home screen. Open it from there from now on.

2 MIN

26

FUN!

Add kids, shows and times

Each kid becomes a channel. Add YouTube channels, playlists or videos to the library, then put them in time slots. Want the other parent along? Invite them from the app, and they join only when they tap **Join**.

> **YOU SHOULD SEE:** a week of programmes. This is the real work, and the best part.

15 MIN

27

ON AIR!

Link a TV

Install the TV app first. **The Samsung guide** or **the Android TV guide** walks you through it. Open it on the TV and it shows a pairing code and a QR code. On your phone, go to **Your TVs** › + **Link a TV** and type the code.

> **YOU SHOULD SEE:** the TV tune in to your channels. After that it checks for your changes every 30 seconds.

2 MIN

PART 6 · KEEPING IT RUNNING *The odd Saturday of upkeep*

U1

Updates

When a new version comes out, fetch it and deploy again. Your settings and secrets stay put. Tip: copy your edited `wrangler.jsonc` to `cloud/wrangler.local.jsonc` once; git ignores that file, every `npm run` command uses it, and updates never clash with your values. The deploy also applies any new database changes.

TYPE THIS, FROM THE SATURDAY-MORNINGS FOLDER

```
git pull
cd cloud
npm install
npm run deploy
```

YOU SHOULD SEE: "Deployed saturday-morning" again. If the update changed anything in `apps/tv-samsung` or `apps/tv/js/config.js`, rebuild and reinstall the TV app too.

EVERY UPDATE · 10 MIN

U2

Back up the database

Everything your family set up is in D1. Now and then, save a copy to your computer as one file. Keep it somewhere private, because it has your family's emails in it.

TYPE THIS

```
npx wrangler d1 export saturday-morning --remote --output=backup.sql
```

YOU SHOULD SEE: a new file, `backup.sql`, in the `cloud` folder. Rename it with today's date.

MONTHLY · 2 MIN

U3

The daily job

Every day at 09:17 UTC the station refreshes YouTube titles older than 20 days and deletes YouTube data older than 30 days, as YouTube's rules require. It runs by itself. To watch it work on your computer:

TYPE THIS

```
npx wrangler dev --test-scheduled
```

YOU SHOULD SEE: a local address; open `/__scheduled` on it and the job runs once. In the live station, the Worker's **LOGS** in the Cloudflare dashboard show each run.

NOTHING TO DO

U4

Moving to a new address

Each TV app is built with your station's address inside it. Move the station and you rebuild the TV app and link the TV again. Keep the old address serving (leave `"workers_dev": true`) until every TV has moved, and don't redirect it, because old TV apps only trust the address they were built with.

> **YOU SHOULD SEE:** under **YOUR TVS**, when each TV last checked in. When they're all on the new address, you're done.

RARELY · 30 MIN

U5

What it costs

Nothing, for a household. Cloudflare's free plan covers the Worker, the database and the daily job comfortably. Resend's free tier sends far more sign-in codes than a family needs. YouTube's 10,000 daily units are free. The only bill you might pay is a domain, about $10 a year. (Free tiers change now and then; check each provider's pricing page if your station grows into a street.)

$0

PART 7 · OR: ONE COMPUTER AT HOME

No cloud, no accounts, just a look

H1

QUICK LOOK

Run it on a computer in the house

For a quick look without Cloudflare, the original home server runs on your own computer. No sign-in, no email and no YouTube key (it looks videos and playlists up by their links). The computer has to stay on, though, and it only works on your home Wi-Fi. Your family's data lives in one file, `home-server/data/household.json`.

TYPE THIS, FROM THE SATURDAY-MORNINGS FOLDER (NOT CLOUD)

```
node home-server/server.js
```

Find your computer's IP address (on a Mac: **System Settings › Wi-Fi › Details**; it looks like `192.168.1.20`). On your phone, open `http://192.168.1.20:8790/setup/` with your own numbers. For the TV, put `TV_SERVER=http://192.168.1.20:8790` in `apps/tv-samsung/.build-tv.local` before you build the TV app.

> **YOU SHOULD SEE:** the setup page on your phone, and the TV app finding the computer when it opens.

15 MIN

STUCK?

No sign-in code arrives	Check spam. Still using `onboarding@resend.dev`? Then only your own Resend email gets codes, so do step 23. Run `npx wrangler tail` while you sign in and look for `resend failed`.
"Not approved" or a waiting-list message	`EARLY_ACCESS` is `"on"`. Add your email to `ADMIN_EMAILS` and deploy, or set it to `"off"`.
Deploy says the database isn't found	The `database_id` in `wrangler.jsonc` doesn't match step 11, or the database isn't named `saturday-morning`. `npx wrangler d1 list` shows yours.
Deploy says "not authorized"	Run `npx wrangler login` again. Check `routes` doesn't still point at our domain (step 13).
Channel links or search fail	The YouTube key is missing, not enabled for YouTube Data API v3, or over today's quota. Check step 8, then try tomorrow.
The passkey won't work	`RP_ID` must be the domain you're on (or its parent). Moved address? Make a new passkey; email codes always work.
The settings file won't save, or deploy complains about it	A missing quote or comma. Compare your lines with the boxes above, character by character.
The TV shows a black screen or "Can't connect right now"	The TV app has the wrong address in it. See the Samsung guide's Stuck box.

Still stuck? Open an issue on the code's GitHub page with the command you ran and exactly what it said. Remove any keys first.

That's a station. *Next stop: the Samsung TV guide or the Android TV guide. Or, if this made you tired just reading it, the home-delivery edition does all of it for you. You sign in, type the code your TV shows, and* **enjoy your own Saturday mornings.**


Your station
is ready for broadcast
Please stand by for the TV.